

Bayu Febriansyah

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EDUCATION

University of Illinois Urbana-Champaign

Expected May 2027

M.S. in Aerospace Engineering; Current GPA: 3.83/4.00

Urbana, Illinois

Bandung Institute of Technology (ITB)

Apr 2023

B.S. in Aerospace Engineering; Cum Laude; GPA: 3.64/4.00

Bandung, Indonesia

EXPERIENCE

3D Printing Engineer

May 2026 – Present

Illinois MakerLab

Champaign, Illinois

- Designed, optimized, and manufactured mechanical components, translating requirements into CAD models and functional parts; refined geometry, assembly, and printing parameters to improve manufacturability and reliability.

Undergraduate Researcher

Oct 2022 – Apr 2023

Lightweight Structures Laboratory

Bandung, Indonesia

- Developed and validated nonlinear finite element models in LS-DYNA; performed parametric and sensitivity analyses to characterize collapse mechanisms, energy absorption, and design tradeoffs in lightweight, thin-walled structures.

Flight Data Engineering Intern

Jul 2022 – Sep 2022

PT Aering

Bandung, Indonesia

- Designed and implemented a Python flight-reconstruction workflow, integrating coordinate correction, altitude and heading processing, and geospatial validation to generate X-Plane simulation input with positional accuracy within 8 m.

Mechanical Engineering Intern

Jun 2021 – Aug 2021

Gexcon

Jakarta, Indonesia

- Conducted ANSYS CFD simulations to characterize fire behavior and propagation under industrial plant accident scenarios, supporting hazard evaluation, plant risk assessment, and safety engineering consulting.

SELECTED PROJECTS

Carbon-Epoxy Composite I-Beam | Baur Research Group, UIUC

2026

- Led fabrication from the team's numerical design, adapting layup to prepreg, tooling, and equipment constraints; replaced foam-filled Nomex with aluminum honeycomb because uniform foam injection proved impractical.

Ultrathin Bistable Composite Boom | Space Structures Laboratory, UIUC

2026

- Built MATLAB laminate stiffness and strain-energy models; screened 5,476 cases, identifying tradeoffs between transition energy and coiled geometry. Fabricated a boom and compared predicted stable shapes with DIC curvature.

Medical UAV System Architecture | Systems Engineering Course, UIUC

2025

- Developed ConOps, requirements, functional decomposition and interface matrix for proposed medical UAV logistics: mission planning, airspace coordination, telemetry monitoring, fleet management and safety intervention.

Impact Analysis of Functionally Graded Crash Boxes | Undergraduate Thesis, ITB

2023

- Compared five taper ratios under axial impact. Identified a graded-thickness multicell design with simulated specific energy absorption 124% above a conventional square tube and 11.8% above a conventional multicell tube.

TECHNICAL SKILLS

Programming and Data Processing: Python, C, MATLAB, Simulink, Mathematica, LaTeX, Microsoft Office.

CAD and Design: Siemens NX, SOLIDWORKS, Fusion 360, AutoCAD, PrusaSlicer, OrcaSlicer, Bambu Studio, Cura.

Engineering Simulation and Analysis: ANSYS, LS-DYNA, Abaqus, HyperMesh, Patran/Nastran, JavaFoil.

Languages: Indonesian (native) and English (fluent, TOEFL iBT 109/120, IELTS 8.0/9.0).

HONORS, AWARDS, AND ACTIVITIES

Tau Beta Pi Engineering Honor Society (2026): Member; recognizes academic excellence and exemplary character.

Indonesia Endowment Fund for Education (LPDP) (2025): Awardee; Indonesian government graduate scholarship.

Dean's List, Bandung Institute of Technology (2023): Awardee; ranked among the top 5% of students consistently.

Memberships: Society for the Advancement of Material and Process Engineering (SAMPE), UIUC Chapter; American Institute of Aeronautics and Astronautics (AIAA), UIUC Chapter; Aerospace Graduate Student Advisory Committee (AeroGSAC); Indonesian Students Association (PERMIAS), UIUC; Indonesian Aerospace Engineering Alumni Association (IAP ITB).